Yuru Sun

Email: yuru.sun@monash.edu | Website: yurusun.netlify.app/

Summary

PhD in Financial Econometrics at Monash University with expertise in asset pricing, empirical finance, statistical modelling, and applied quantitative research. Experienced in applying econometric and statistical methods to large financial datasets to evaluate market behaviour, regulatory impacts, and risk management. Strong programming capability in R, Python, MATLAB and Excel, with experience communicating complex quantitative findings to both technical and non-technical stakeholders.

Work Experience

Teaching Associate *Monash Business School*, Feb 2023 - Present

- Delivered undergraduate and postgraduate teaching in statistics, econometrics, forecasting, and data analytics, communicating complex quantitative concepts to diverse audiences.
- Coordinated tutorial teams and unit delivery as a head tutor across multiple units.
- Recipient of the Excellent Teaching Team Awards (2024, 2025), recognising excellence in teaching and collaboration.

Research Assistant (Modeller) *Monash Business School*, Aug 2023 - Oct 2024

- Conducted economic and financial market analysis using high-frequency bond market data to evaluate the impact of regulatory and policy changes.
- Developed reproducible R workflows to clean and analyse financial market data, producing insights through visualisation and statistical analysis to support policy evaluation and economic research.
- Implemented econometric models, bootstrap inference, hypothesis tests on jumps around policy changes and robustness analysis to analyse policy impacts and translate statistical findings into clear and decision-oriented insights.

Research Assistant (Data Analyst) *Melbourne Business School*, Aug 2022 - Jan 2023

- Automated large-scale high-frequency equity S&P 500 constituent equity data retrieval from a commercial database (Refinitive) and processing pipelines using Python and R for millisecond returns of stocks, improving efficiency and reproducibility of dataset construction;
- Cleaned and validated high-frequency time-series data, implemented quality checks, and produced analysis-ready datasets to support modelling and forecasting tasks;
- Collaborated with academic stakeholders to translate project requirements into structured datasets and deliverables.

Macroeconomics Research Analyst Internship *Liaowang Think Tank*, July 2021 - Oct, 2021

- Conducted macroeconomic research on fiscal and monetary policy, evaluating their implications for economic activity.
- Produced written research reports and presentations, communicating economic insights to non-technical stakeholders.

Research Project Experience

Improvements on Option Implied Market Risk Measures *Monash,* Nov 2024 - Present

- Led the project in collaboration with external academics, overseeing model development and empirical analysis.
- Developed forward-looking asymmetric systematic risk measures from option prices using option pricing theory and numerical optimisation techniques to investigate the pricing of downside and upside systematic risk in the cross-section of equity returns.
- Processed and analysed large-scale equity return and OptionMetrics option datasets in R and MATLAB, and designed simulation studies to evaluate model performance and robustness under varying market conditions.

Modelling Financial Price Jumps *UniMelb & Monash,* July 2022 - 2025

- Developed Bayesian nonparametric (Dirichlet process mixture), state space and Markov switching models to analyse equity and commodity market dynamics, price jumps, latent market states and uncertainty under varying market conditions with probabilistic forecasting.
- Processed and analysed high-frequency data in R, Python and MATLAB, building predictive analytics and forecasting pipelines to assess market risks, return distributions and investment outcomes.
- Implemented reliable version control workflows using Git and executed large-scale computational jobs on HPC clusters via SLURM.
- Communicated complex quantitative results through clear visualisations, conference presentations, and technical reports.

Optimal Probabilistic Forecast for Risk Management *Monash,* March 2020 - 2022

- Developed an advanced statistical model to analyse financial market behaviour, especially equity indices, and generate probabilistic forecasts of future returns.
- Applied probabilistic forecasts of returns to risk management through Value-at-Risk (VaR) analysis, Expected Shortfall estimation and developed hedging strategies with VIX index and futures.

Education

2023 - 2026	PhD in Financial Econometrics - **Monash University (Transferred from UniMelb)**
2021	Graduate Certificate in Economics Analysis at **University of Sydney**
2017 - 2020	Bachelor in Commerce (Finance and Econometrics, 1st class Honours) at **Monash University**

Skills

R, MATLAB, VBA, Python, Linux, Git, Excel, STATA, SQL, Refinitive, OptionMetrics

Honours & Awards

2024, 2025	Excellent Teaching Team Awards, **Monash Business School**
2025	Prestigious International Conference Award, **Monash Business School**
2025	CFM/SoFiE Scholarship, **Capital Fund Management & Society for Financial Econometrics**
2024	Best Poster Runner-Up, **Time Series and Forecasting Symposium**, University of Sydney
2023	Co-funded Monash Graduate Scholarship, **Monash University**
2022	Faculty of Business and Economics Graduate Research Scholarship, **University of Melbourne**
2020	Econometrics Memorial Scholarship, **Monash University**
2020	Meritorious Winner, **Interdisciplinary Contest in Modelling (ICM)**